JUAN MANUEL RUIZ MUÑOZ

ML Engineer · Computer Vision & LLMs

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EU citizen (no sponsorship required) ◇ Open to relocation, hybrid or remote (CET)

PROFILE

ML Engineer with a Physics and Mathematics background and 2+ years taking computer vision and language models from research to production. I build end-to-end systems (data, training, evaluation, deployment) for industrial inspection, including a LoRA-fine-tuned multimodal OCR running at 98.56% exact match in production. Rigorous by training: I want to understand why a model works before shipping it. Published at DCAI'25.

EXPERIENCE

Vraia Corp. 06/2025 – Present

ML Engineer · Computer Vision & LLMs *Elche, Spain*

- Built a **multimodal OCR** for tyre-sidewall text end to end: fine-tuned **GLM-OCR** with **LoRA** (LLaMA-Factory) on a 2×RTX 3090 server, added a post-processing correction stage and served it with **vLLM**, reaching **98.56% exact match in production**.
- Designed and trained **detection and segmentation** models (CNNs, Vision Transformers, YOLO, RF-DETR) for industrial tyre inspection and **defect detection**: **0.83 mAP@50:95** on multi-class segmentation over proprietary datasets of thousands of images.
- Diagnosed and mitigated mask collapse on thin ring-shaped parts in a 7-class **RF-DETR** segmentation model (window/shutter components) by re-weighting the mask loss.
- Built a **synthetic defect generation** pipeline with generative models to augment scarce defect classes and improve detection robustness.
- **MLOps**: ETL and augmentation pipelines for visual datasets; deployment and monitoring with **Docker, MLflow** and TensorBoard; evaluation protocols (precision, recall, F1, mAP, IoU); algorithm specifications in **C++** for integration by the client's engineering team.

AIR Institute & BISITE Research Group (University of Salamanca) 06/2024 – 06/2025

Data Scientist · AI Researcher *Salamanca, Spain*

- Co-designed an **LLM + RAG** architecture that builds a verified ontology incrementally to detect false claims in fiction (precision 0.98, F1 0.86); **published at DCAI'25**.
- Implemented **RAG** and semantic-retrieval pipelines (ChromaDB) and **multi-agent** systems (Ollama, DeepSeek, GPT-4o-mini), focused on reducing cost and latency.
- Delivered computer vision projects (detection, classification, segmentation) across the full ML lifecycle, working directly with clients on requirements and on adapting state-of-the-art methods to real use cases.

Miguel Hernández University – SCOLAb Group 01/2024 – 07/2024

Research Intern *Elche, Spain*

- Developed Lagrangian methods for the automatic identification of atmospheric blocking (ERA5, HYSPLIT, DBSCAN); basis of my bachelor's thesis and of a paper in preparation.

SELECTED PROJECT

vcalib – calibration filters for RF-DETR under illumination shift open source

- A differentiable filter with a few thousand parameters that corrects illumination shift *before* a frozen detector, with no retraining and no labels. On 6 real scenes unseen during fitting, **AP@[.5:.95] rises from 0.377 to 0.580** with the filter (0.544 under the reference condition), measured against independent SAM 3 pseudo-ground truth. Fitted in ∼3.5 min on a laptop CPU.

EDUCATION

BSc in Physics – University of Alicante 2020 – 2024

- 120+ ECTS of the BSc in Mathematics taken in parallel (UA simultaneous-degree programme). SICUE exchange scholarship, University of Salamanca (2022 – 2023).

PUBLICATIONS

- **DCAI'25**: *AI-Powered Ontology-Based Architecture for Misinformation Detection in Fiction Works.* [PDF]

SKILLS

LLMs & NLP	LoRA fine-tuning (LLaMA-Factory), Transformers, RAG, vLLM, Ollama, vector DBs (ChromaDB, FAISS), multi-agent
Computer Vision	PyTorch, OpenCV, CNNs, ViTs, YOLO, RF-DETR, SAM, generative models, detection, segmentation
MLOps & Code	Python, C++ (spec), R, Docker, MLflow, TensorBoard, Git, AWS, Linux, uv; statistical analysis
Languages	Spanish (native) · English (working proficiency)